

Multispectral Remote Sensing and Machine Learning-Based Land Use Land Cover (LULC) Characterization of Shariatpur District, Bangladesh

1. Study Area Profile

Shariatpur District is situated within the central-southern deltaic region of Bangladesh (Dhaka Division), bounded by the active Padma River to the north and east, Madaripur District to the west, and Barishal and Chandpur districts to the south. Geographically, it extends approximately from 23°06'00" N to 23°25'30" N in latitude and 90°10'00" E to 90°40'00" E in longitude. The district is characterized by flat, low-lying alluvial floodplain topography formed by the dynamic depositional and erosional processes of the Ganges-Brahmaputra-Meghna delta. Because of its direct frontage along the braided Padma river channel, Shariatpur features an intricate hydrological network of secondary rivers, canals, seasonal wetlands, and continuously shifting sandy charland formations, making its landscape highly sensitive to seasonal flooding, riverbank erosion, and intense agricultural cultivation.

2. True Color Optical Baseline and Landscape Overview

Optical satellite remote sensing in standard natural color composite (Red, Green, Blue) provides the foundational baseline for interpreting physical surface features, hydrological courses, and regional vegetation patterns. The multispectral optical composite of Shariatpur District delineates the primary fluvial geomorphology of the Lower Padma River along its northern boundary, including dynamic sandy charlands, braided channels, and inland agricultural plains. The tonal variations distinguish active sediment-laden waterbodies from terrestrial crop canopy, settlement clusters, and moist alluvial depressions.

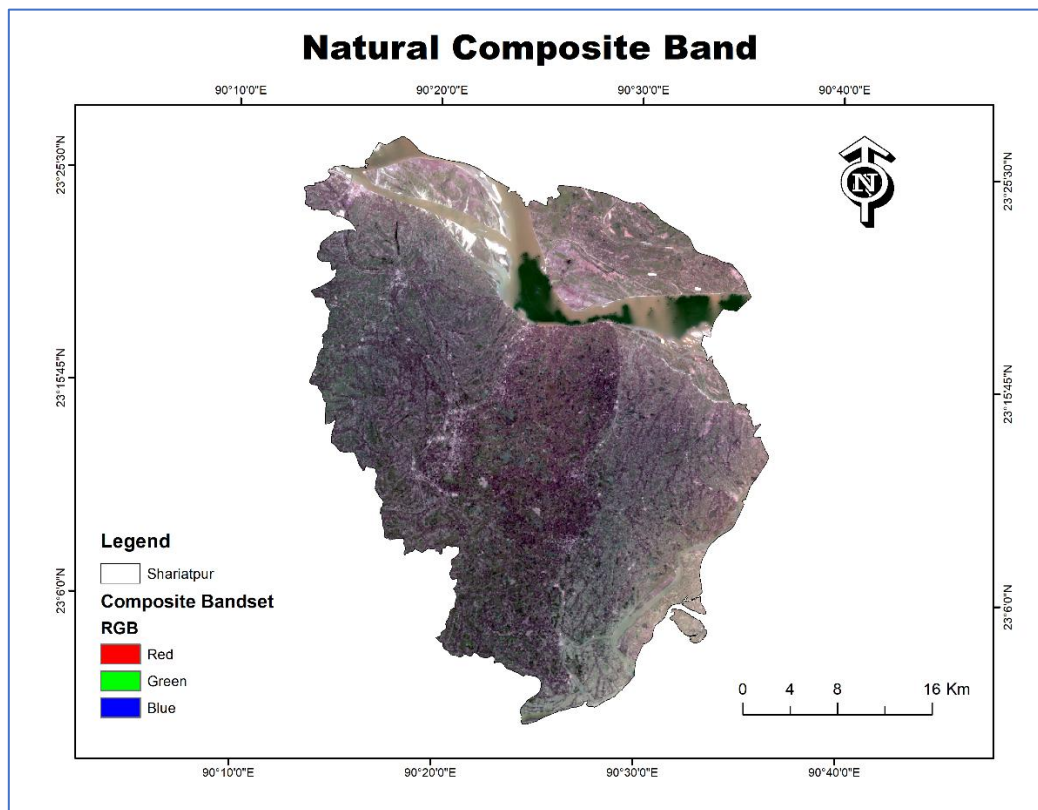


Figure 1: Natural Composite

3. Spectral Index Derivation and Biophysical Surface Modeling

To extract quantitative surface characteristics and separate overlapping spectral signatures prior to final classification, seven distinct biophysical indices were computed: NDVI, SAVI, NDWI, MNDWI, NDBI, BSI, and NDMI.

3.1 Normalized Difference Vegetation Index (NDVI)

NDVI quantifies photosynthetic activity and vegetation vigor by exploiting the contrast between the near-infrared and red spectral bands. In Shariatpur, high NDVI values (dark green) dominate the central and southern rural belts, representing dense homestead agroforestry, orchard canopies, and standing agricultural crops. Moderate values (yellowish-green) represent seasonal fallow fields

and transitional vegetation, whereas low values (dark purple) delineate the deep river waters and non-vegetated sandbars along the Padma corridor.

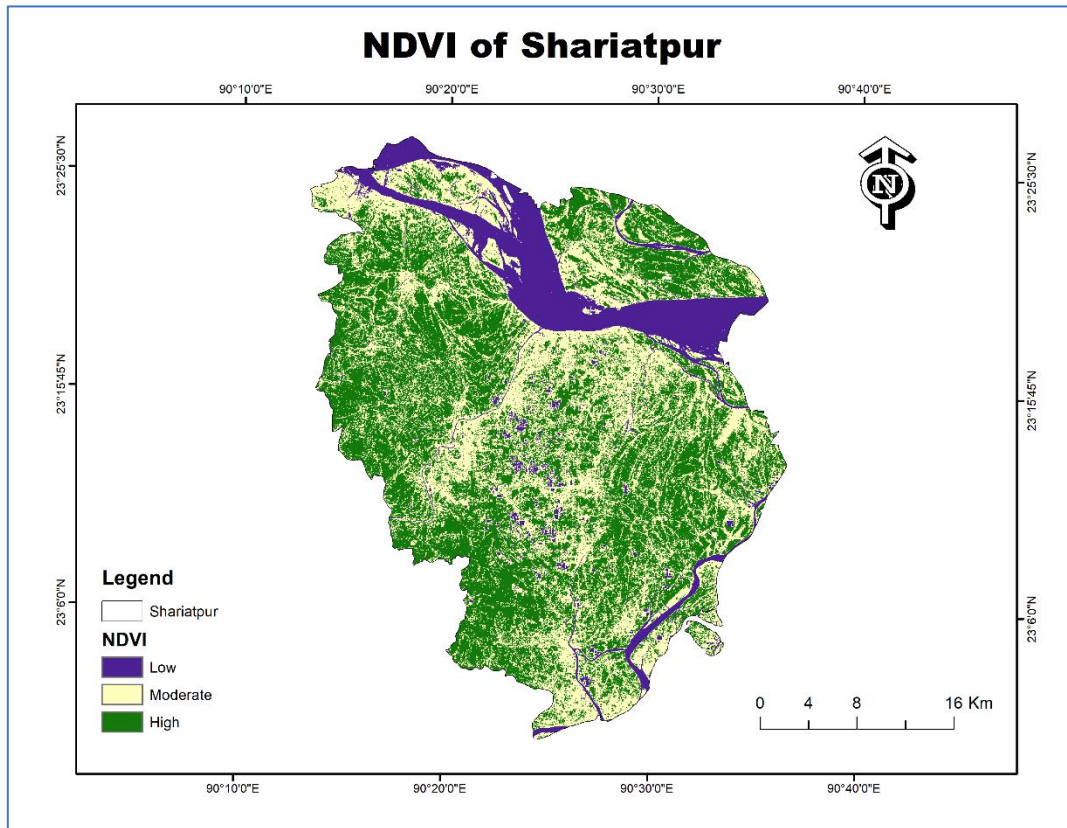


Figure 2: Normalized Difference Vegetation Index (NDVI) Surface of Shariatpur District.

3.2 Soil-Adjusted Vegetation Index (SAVI)

In areas where sparse canopy allows underlying bare soil to influence canopy reflectance, the Soil-Adjusted Vegetation Index incorporates a soil-brightness correction factor ($L = 0.5$) to minimize soil background interference. The SAVI map exhibits a strong gradient where high values (dark olive green) capture true vegetated biomass across rural homesteads, moderate values represent cultivated croplands, and low values (light green) isolate open river sands, bare charlands, and active water surfaces.

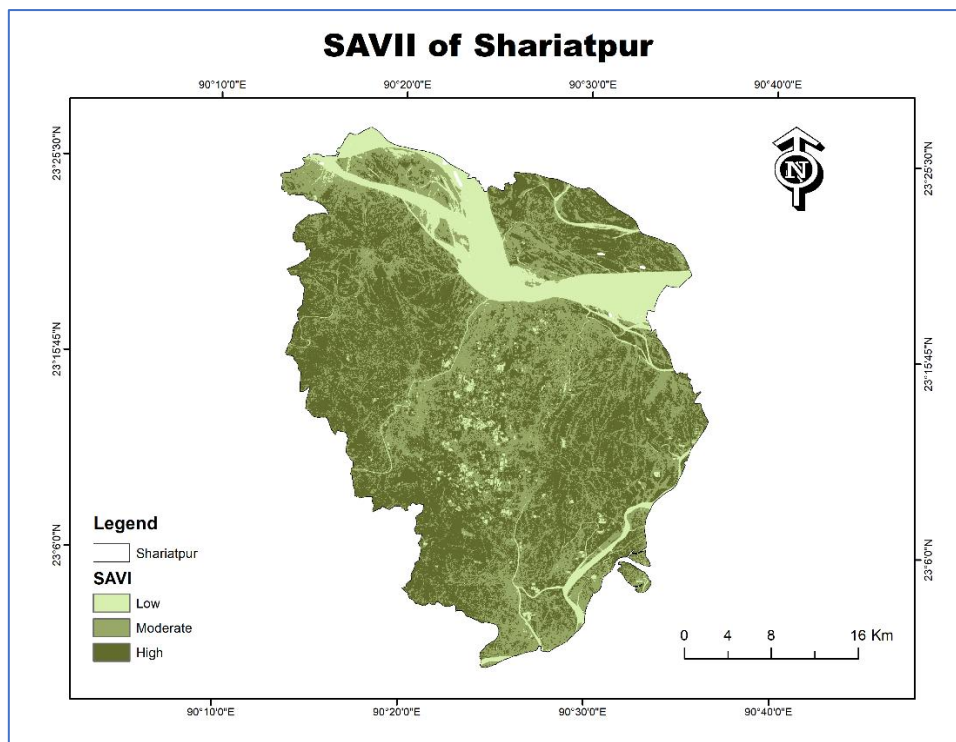


Figure 3: Soil-Adjusted Vegetation Index (SAVI) Distribution of Shariatpur District.

3.3 Normalized Difference Water Index (NDWI)

NDWI utilizes the green and near-infrared reflectance bands to delineate open surface water features by suppressing soil and terrestrial vegetation noise. High NDWI values (dark blue) clearly map the primary Padma River channel and perennial tributary networks. Moderate NDWI values identify intermediate wetland zones, low-lying beels, and aquaculture ponds distributed across the rural interior, while low values delineate dryland agriculture and settlement areas.

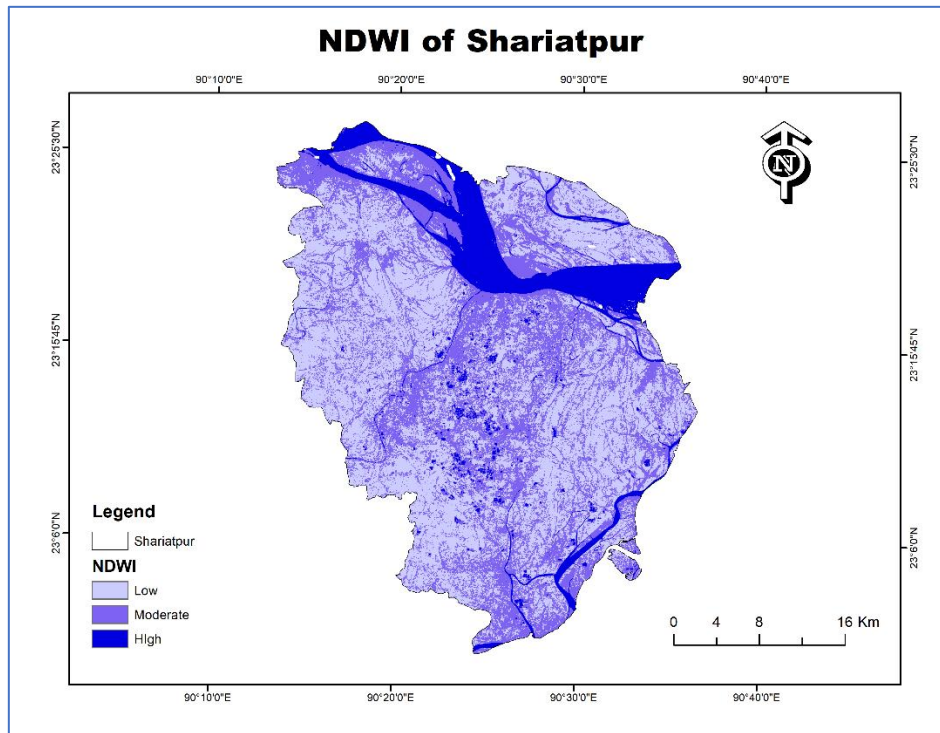


Figure 4: Normalized Difference Water Index (NDWI) Map of Shariatpur District.

3.4 Modified Normalized Difference Water Index (MNDWI)

MNDWI enhances water feature delineation in built-up and mixed rural environments by replacing the near-infrared band with short-wave infrared (SWIR). This modification eliminates false water classifications caused by urban infrastructure and saturated soils. In Shariatpur, the high MNDWI tier (dark blue) accurately isolates deep open waters, while moderate values (cyan) pinpoint saturated agricultural depressions, secondary drainage channels, and intertidal mud flats.

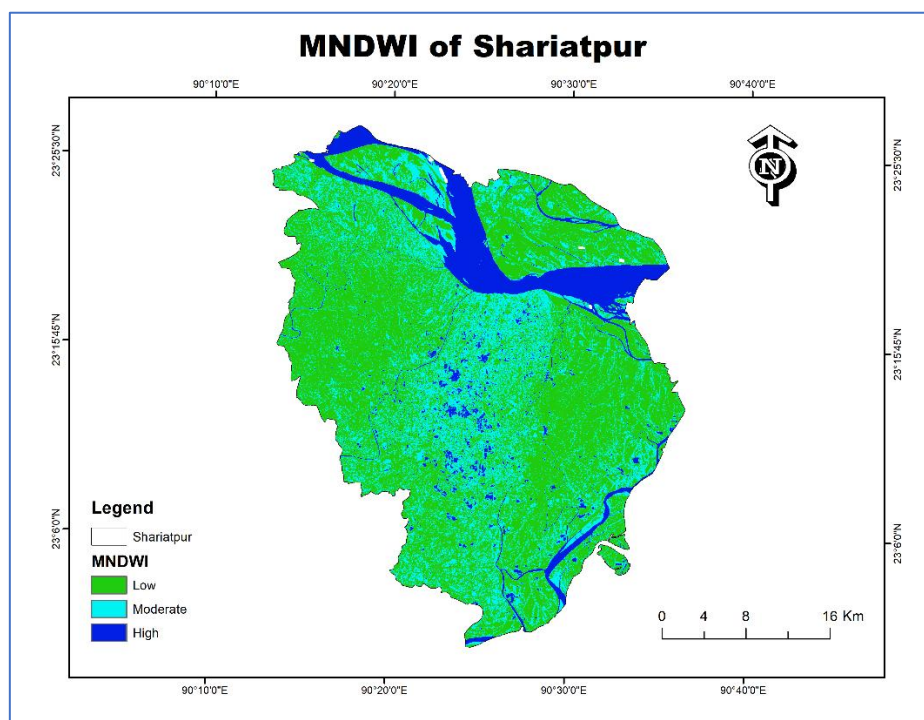


Figure 5: Modified Normalized Difference Water Index (MNDWI) Map of Shariatpur District.

3.5 Normalized Difference Built-Up Index (NDBI)

NDBI leverages the high reflectance of man-made impervious surfaces in the SWIR spectrum compared to NIR. High NDBI zones (dark red/brown) map out dense municipal settlements, commercial centers, concrete transport infrastructure, and dry bare char surfaces. Moderate values represent rural homestead clusters interspersed with trees, while low values correspond to water bodies and healthy vegetated cropland.

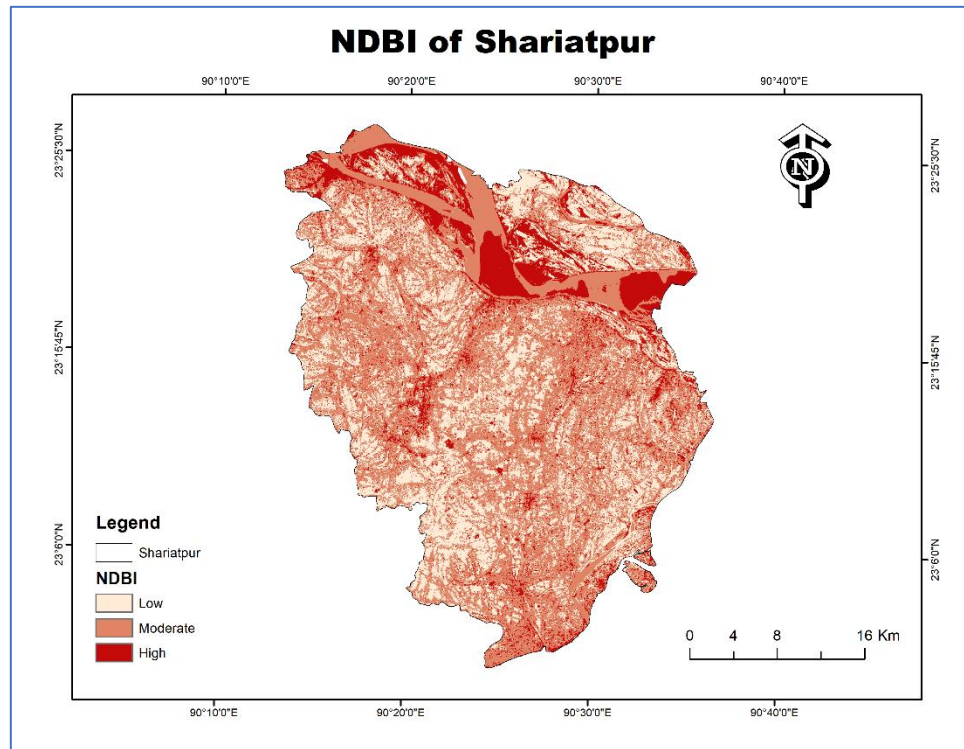


Figure 6: Normalized Difference Built-Up Index (NDBI) Map of Shariatpur District.

3.6 Bare Soil Index (BSI)

BSI integrates blue, red, NIR, and SWIR bands to map bare soil, agricultural fallow land, unpaved roads, and exposed riverbed sand. High BSI areas (dark brown/red) are concentrated along the northern Padma riverbed and newly accreted sandbars (chars). Moderate values (orange) span active agricultural plots between cropping cycles, while low values represent water bodies and dense canopy cover.

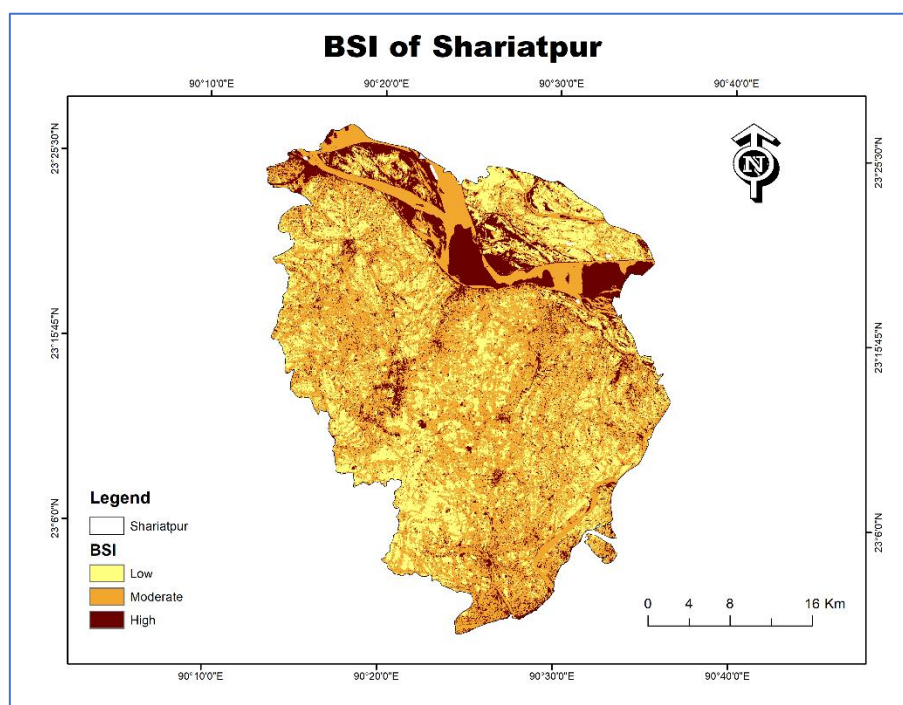


Figure 7: Bare Soil Index (BSI) Map of Shariatpur District.

3.7 Normalized Difference Moisture Index (NDMI)

NDMI evaluates canopy water content and soil moisture levels using NIR and SWIR spectral bands. High NDMI values (dark blue) highlight dense, well-hydrated vegetation, inland wetlands, and canal networks. Moderate values correspond to rainfed agricultural land, while low values (light blue/grey) delineate dried sandy riverbeds, bare charlands, and built structures where moisture retention is minimal.

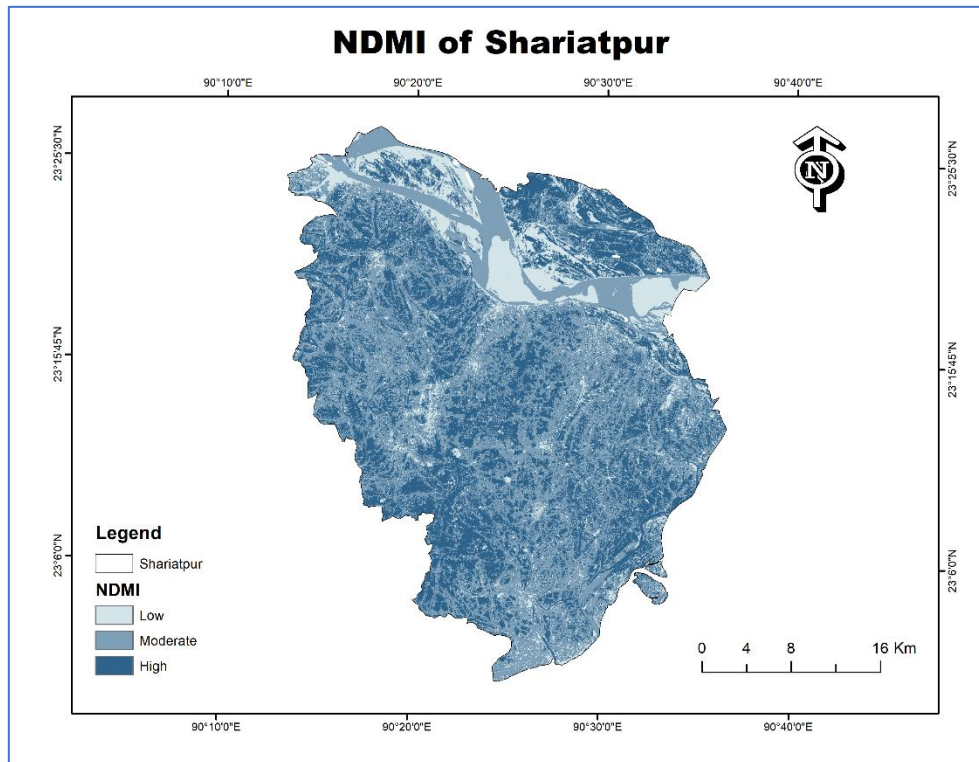


Figure 8: Normalized Difference Moisture Index (NDMI) Map of Shariatpur District.

4. Land Use Land Cover (LULC) Classification Analysis

By integrating multispectral bands with derived biophysical indices into a supervised machine learning classification workflow, the landscape of Shariatpur was categorized into five distinct LULC classes: Waterbody, Agriculture, Vegetation, Built-Up Area, and Barren Land.

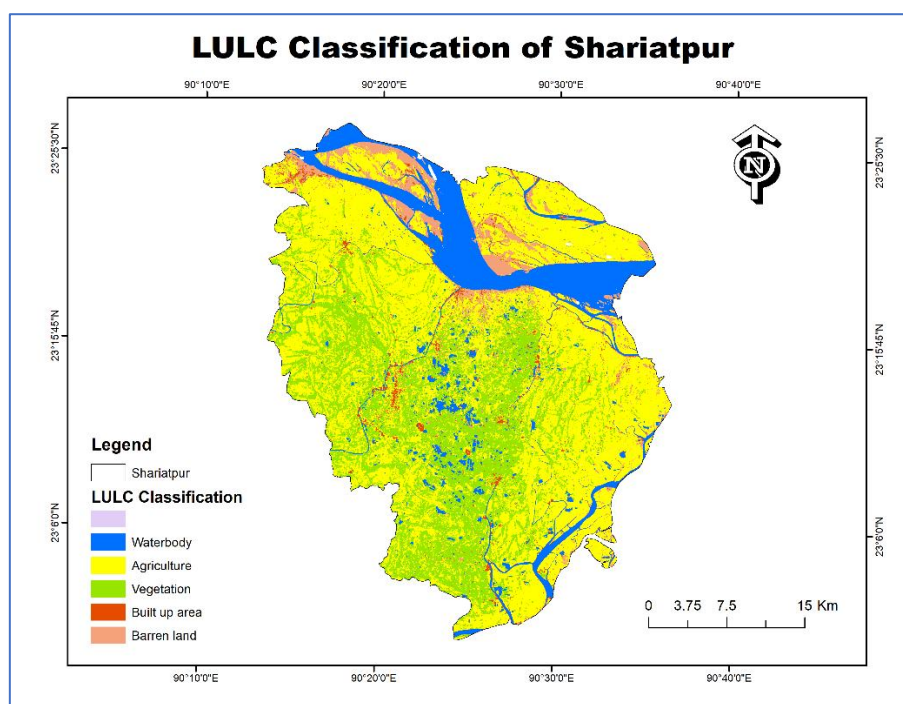


Figure 9: Supervised Land Use Land Cover (LULC) Classification Map of Shariatpur District.

The classification map reveals that Agriculture (Yellow) is the dominant land use class across the district, covering the expansive floodplains in both central and southern regions. Vegetation (Green) forms a dense, continuous network corresponding to homestead forestry, orchards, and rural settlements, particularly clustered in the central-south zone. Waterbody (Blue) covers the northern Padma River, active meandering river loops, and scattered inland ponds. Barren Land (Peach/Orange) is primarily localized along dynamic sandy river chars and eroded riverbanks in the north, while Built-Up Area (Red) forms distinct linear and nodal clusters marking municipal centers, highway networks, and rural bazaars.

5. Synthesis and Strategic Conclusion

The integration of multispectral indices (NDVI, SAVI, NDWI, MNDWI, NDBI, NDMI, BSI) provides a robust feature space for classifying complex deltaic landscapes. In Shariatpur, the dynamic interaction between river channel morphometry, extensive agricultural cultivation, and expanding rural settlements requires continuous remote sensing surveillance. The generated LULC classification offers critical baseline intelligence for regional land-use zoning, floodplain risk management, agricultural monitoring, and infrastructure planning along the Padma river basin.